SEPARATING THE CLOSURE OF THE IMAGE OF A CLOSED SET BY AN ELEMENT CONGRUENT TO ONE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Two closed subsets with empty intersection are defined by ideals generating the unit ideal, and vanishing of a function on a closed set means a power lies in the ideal. Algebra, Lemma \ref{algebra-lemma-Zariski-topology}, cited at proof lines 1950-1951 and 1954-1956. Gives the unit ideal decomposition and the power statement.

Source reference. Original lemma 07M3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9713–9713. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00E0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 517–517. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to S$ be a ring map. Let $I \subset R$ be an ideal of R and let $J \subset S$ be an ideal of S. If the closure of the image of $V(J)$ in $\mathrm{Spec}(R)$ is disjoint from $V(I)$, then there exists an element $f \in R$ which maps to 1 in R/I and to an element of J in S.*

Proof. Let $I' \subset R$ be an ideal such that $V(I')$ is the closure of the image of $V(J)$. Then $V(I) \cap V(I') = \emptyset$ by assumption and hence $I + I' = R$ by Algebra, Lemma 00E0. Write $1 = g + f$ with $g \in I$ and $f \in I'$. We have $V(f') \supset V(J)$ where f'

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is the image of f in S. Hence $(f')^n \in J$ for some n, see Algebra, Lemma 00E0. Replacing f by f^n we win. $\square$